In [35]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import matplotlib.pyplot as plt
from scipy.stats import skew
from scipy.stats.stats import pearsonr
import sklearn as sk
from scipy import stats
from sklearn import preprocessing
from sklearn.preprocessing import LabelEncoder
# novelty analysis
from sklearn.neighbors import LocalOutlierFactor
from sklearn import datasets, linear_model
from sklearn.model_selection import train_test_split
from sklearn import model_selection
from sklearn import metrics
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix
from sklearn.metrics import accuracy_score
```

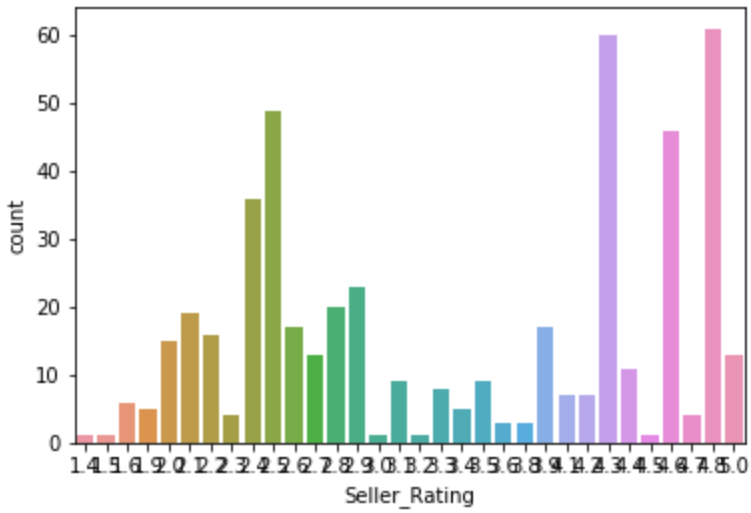
In [36]:
```
df = pd.read_excel ('Mobile info.xlsx') #for an earlier version of Excel, yo
```

In [37]:
```
df.head(4)
```

Out[37]:

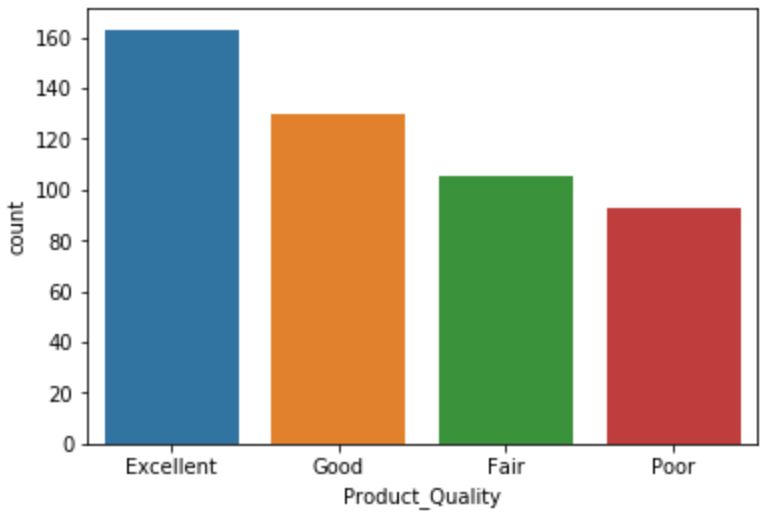
	Product_Name	Seller_Name	Overall_Rating	Overall_Review	Seller_Rating	A
0	RedmiNote6Pro	FlashstarCommerce	352708	32857	4.7	
1	RedmiNote6Pro	SuperComNet	352708	32857	4.3	
2	RedmiNote7	SuperComNet	10387	1057	4.3	
3	RedmiY1	samaymobiles	17421	1634	3.1	

In [43]:
```
sns.countplot(df['Seller_Rating'],label="Count")
plt.figure(figsize=(15,9))
plt.savefig('M_Seller')
plt.show()
```




```
<Figure size 1080x648 with 0 Axes>
```

In [33]:
```
sns.countplot(df['Product_Quality'],label="Count")
plt.savefig('Product Quality')
plt.show()
```



In [34]:
```
labelencoder = LabelEncoder()
df['Product_Name'] = labelencoder.fit_transform(df['Product_Name'])
df['Seller_Name'] = labelencoder.fit_transform(df['Seller_Name'])
df['Product_Quality'] = labelencoder.fit_transform(df['Product_Quality'])
```

In [7]:
```
df['Seller_Rating']= df['Seller_Rating'].astype('category')
df['Seller_Rating'] = df['Seller_Rating'].cat.codes
df['Average_Rating']= df['Average_Rating'].astype('category')
df['Average_Rating'] = df['Average_Rating'].cat.codes
df['Overall_Rating']= df['Overall_Rating'].astype('category')
df['Overall_Rating'] = df['Overall_Rating'].cat.codes
df['Overall_Review']= df['Overall_Review'].astype('category')
df['Overall_Review'] = df['Overall_Review'].cat.codes
```

In [8]:
```
df.head()
```

Out[8]:

	Product_Name	Seller_Name	Overall_Rating	Overall_Review	Seller_Rating	Averag
0	11	16	96	87	29	
1	77	7	96	87	25	
2	78	7	62	40	25	
3	79	14	71	45	15	
4	11	16	89	75	25	

In [9]:

```
feature_cols = ['Seller_Name', 'Seller_Rating', 'Overall_Review', 'Overall_F
X = df[feature_cols] # Features
y = df.Product_Quality # Target variable
```

In [10]:

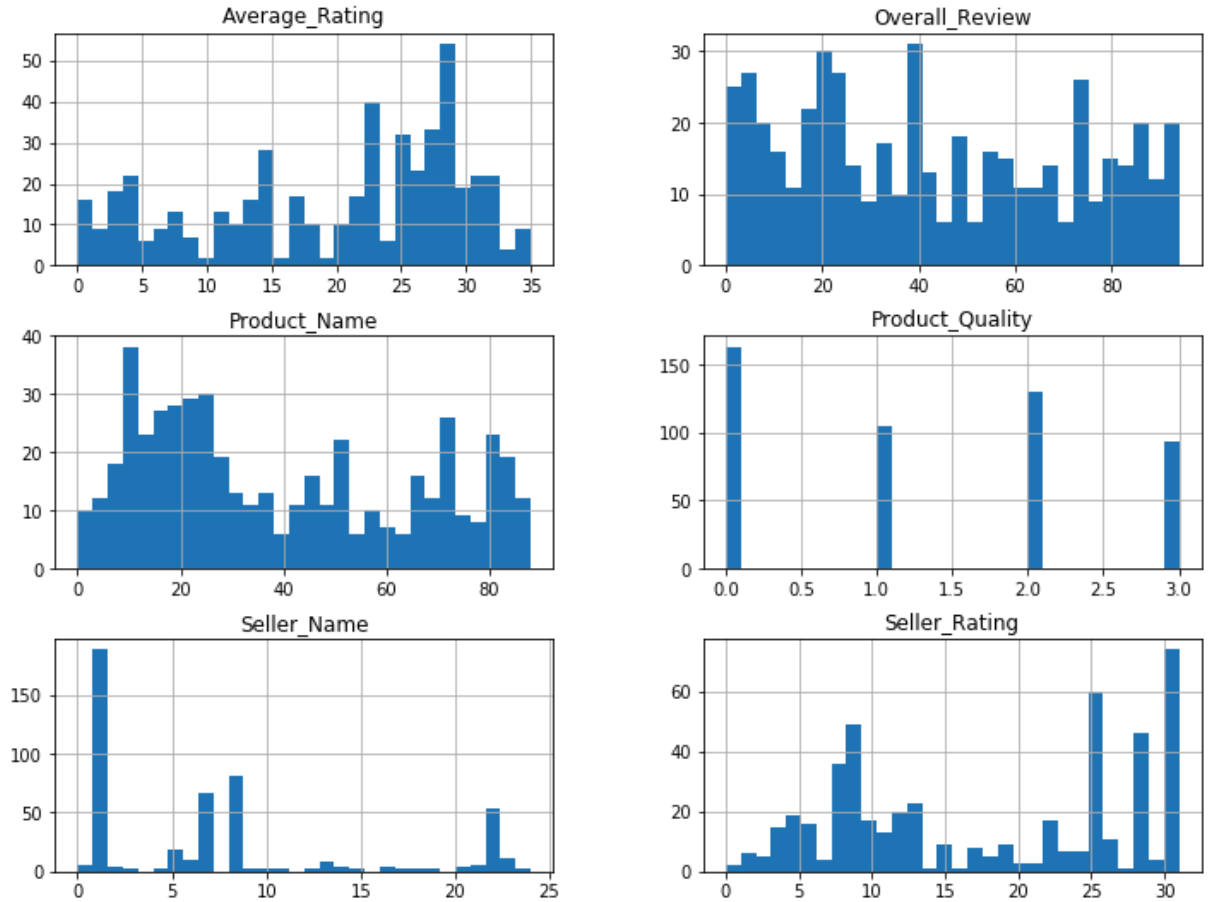
```
df.head(2)
```

Out[10]:

	Product_Name	Seller_Name	Overall_Rating	Overall_Review	Seller_Rating	Averag
0	11	16	96	87	29	
1	77	7	96	87	25	

In [11]:

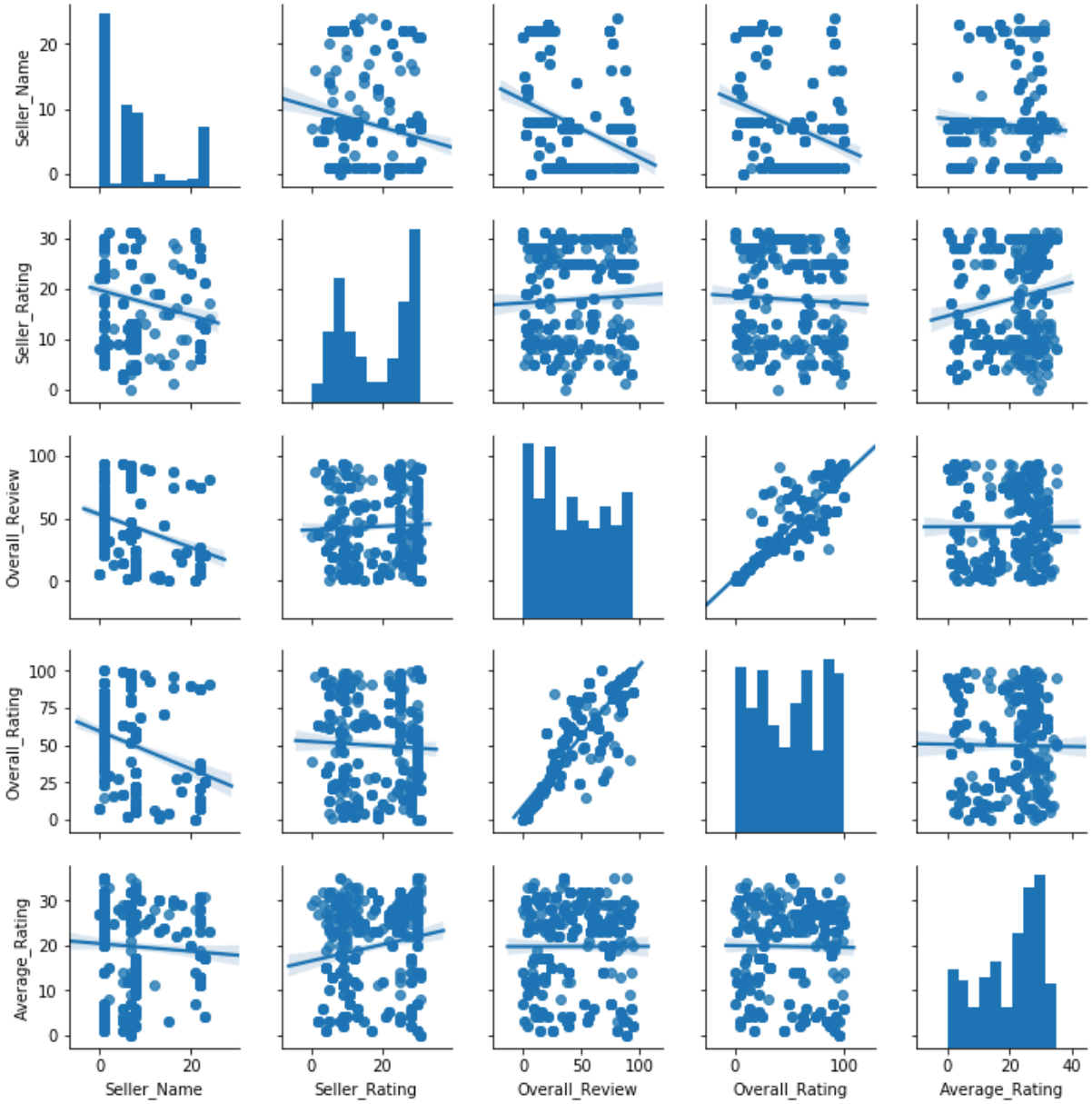
```
import pylab as pl
df.drop('Overall_Rating' ,axis=1).hist(bins=30, figsize=(12,9))
plt.savefig('M_Review')
plt.show()
```



In [12]:
```
import seaborn as sns
import matplotlib.pyplot as plt
```

In [13]:
```
sns.pairplot(X,kind='reg',height=2)
plt.show()
plt.savefig('Mobile_correlation among different attributes')
```

```
C:\ProgramData\Anaconda3\lib\site-packages\scipy\stats\stats.py:1713: Future
Warning: Using a non-tuple sequence for multidimensional indexing is depreca
ted; use `arr[tuple(seq)]` instead of `arr[seq]`. In the future this will be
interpreted as an array index, `arr[np.array(seq)]`, which will result eithe
r in an error or a different result.
  return np.add.reduce(sorted[indexer] * weights, axis=axis) / sumval
```




```
<Figure size 432x288 with 0 Axes>
```

In [14]:
```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.30, ra
```

In [15]:
```
from sklearn import tree
from sklearn.tree import DecisionTreeClassifier

# Create Decision Tree classifer object
clf = DecisionTreeClassifier()

# Train Decision Tree Classifer
clf = clf.fit(X_train,y_train)

#Predict the response for test dataset
y_pred = clf.predict(X_test)
print("Accuracy:",metrics.accuracy_score(y_test, y_pred))
results=confusion_matrix(y_test,y_pred)
print('confusion matrix')
```

```
print(results)
print (classification_report(y_test,y_pred))
```

```
Accuracy: 1.0
confusion matrix
[[49  0  0  0]
 [ 0 36  0  0]
 [ 0  0 39  0]
 [ 0  0  0 24]]
             precision    recall  f1-score   support

          0       1.00      1.00      1.00        49
          1       1.00      1.00      1.00        36
          2       1.00      1.00      1.00        39
          3       1.00      1.00      1.00        24

avg / total       1.00      1.00      1.00       148
```

In [16]:
```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.30, ra
```

In [17]:
```
from sklearn.ensemble import RandomForestClassifier

#Create a Gaussian Classifier
clf=RandomForestClassifier(n_estimators=100)

#Train the model using the training sets y_pred=clf.predict(X_test)
clf.fit(X_train,y_train)

y_pred = clf.predict(X_test)
print("Accuracy:",metrics.accuracy_score(y_test, y_pred))
results=confusion_matrix(y_test,y_pred)
print('confusion matrix')
print(results)
print (classification_report(y_test,y_pred))
```

```
C:\ProgramData\Anaconda3\lib\site-packages\sklearn\ensemble\weight_boosting.
py:29: DeprecationWarning: numpy.core.umath_tests is an internal NumPy modul
e and should not be imported. It will be removed in a future NumPy release.
  from numpy.core.umath_tests import inner1d
Accuracy: 0.9932432432432432
confusion matrix
[[49  0  0  0]
 [ 0 36  0  0]
 [ 0  1 38  0]
 [ 0  0  0 24]]
             precision    recall  f1-score   support

          0       1.00      1.00      1.00        49
          1       0.97      1.00      0.99        36
          2       1.00      0.97      0.99        39
          3       1.00      1.00      1.00        24

avg / total       0.99      0.99      0.99       148
```

In [18]:
```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.30, ra
```

In [19]:
```
from sklearn import svm

#Create a svm Classifier
clf = svm.SVC(kernel='rbf' ,gamma='auto') # Linear Kernel

#Train the model using the training sets
clf.fit(X_train, y_train)

#Predict the response for test dataset
y_pred = clf.predict(X_test)
print("Accuracy:",metrics.accuracy_score(y_test, y_pred))
results=confusion_matrix(y_test,y_pred)
print('confusion matrix')
print(results)
print (classification_report(y_test,y_pred))
```

```
Accuracy: 0.8513513513513513
confusion matrix
[[48  0  1  0]
 [ 5 31  0  0]
 [16  0 23  0]
 [ 0  0  0 24]]
             precision    recall  f1-score   support

          0       0.70      0.98      0.81        49
          1       1.00      0.86      0.93        36
          2       0.96      0.59      0.73        39
          3       1.00      1.00      1.00        24

avg / total       0.89      0.85      0.85       148
```

In [20]:
```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.30, ra
```

In [21]:
```
from sklearn.naive_bayes import GaussianNB

#Create a Gaussian Classifier
gnb = GaussianNB()

#Train the model using the training sets
gnb.fit(X_train, y_train)

#Predict the response for test dataset
y_pred = gnb.predict(X_test)
print("Accuracy:",metrics.accuracy_score(y_test, y_pred))
results=confusion_matrix(y_test,y_pred)
print('confusion matrix')
print(results)
print (classification_report(y_test,y_pred))
```

```
Accuracy: 0.9662162162162162
confusion matrix
[[43  0  2  0]
 [ 0 38  0  1]
 [ 2  0 32  0]
 [ 0  0  0 30]]
             precision    recall  f1-score   support

          0       0.96      0.96      0.96        45
          1       1.00      0.97      0.99        39
          2       0.94      0.94      0.94        34
          3       0.97      1.00      0.98        30

avg / total       0.97      0.97      0.97       148
```

In [22]:
```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.30, ra
```

In [23]:
```
#import the KNeighborsClassifier class from sklearn
from sklearn.neighbors import KNeighborsClassifier
#import metric model to check the accuracy
from sklearn import metrics
#Try running from k=1 through 5 and record testing accuracy
k_range = range(1,5)

score = {}

score_list = []
for k in k_range:
        knn = KNeighborsClassifier(n_neighbors=k)
        knn.fit(X_train,y_train)
        y_pred=knn.predict(X_test)
        score[k] = metrics.accuracy_score(y_test,y_pred)
        score_list.append(metrics.accuracy_score(y_test,y_pred))
# Model Accuracy: how often is the classifier correct?
print("Accuracy:",metrics.accuracy_score(y_test, y_pred))
results=confusion_matrix(y_test,y_pred)
print('confusion matrix')
print(results)
print (classification_report(y_test,y_pred))
```

```
Accuracy: 0.8243243243243243
confusion matrix
[[36  0  9  0]
 [ 0 37  1  1]
 [10  2 22  0]
 [ 0  3  0 27]]
             precision    recall  f1-score   support

          0       0.78      0.80      0.79        45
          1       0.88      0.95      0.91        39
          2       0.69      0.65      0.67        34
          3       0.96      0.90      0.93        30

avg / total       0.82      0.82      0.82       148
```

In [ ]: